Exact and recovered drift and diffusion

SSISDE experiment results

23 October 2026

Final Stochastic SINDy fits using noise recovered from each drift method: $dX_t = \widehat{\mu}(X_t)\,dt + \widehat{\sigma}(X_t)\,dW_t$.

Method	Recovered drift $\widehat{\mu}(x)$	Recovered diffusion $\widehat{\sigma}(x)$	α	ρ	Final MSE
Black–Scholes Exact: $\mu(x) = 0.50000\,x$; $\sigma(x) = 0.30000\,x$					
Method 1	$0.50377\,x$	$0.00324 + 0.29576\,x$	0.00039	0.85000	5.24352×10^{-9}
Method 2	$0.50169\,x$	$0.00318 + 0.29582\,x$	0.00039	0.85000	5.19170×10^{-9}
ODE2	$0.50186\,x$	$0.00319 + 0.29581\,x$	0.00039	0.85000	5.19375×10^{-9}
Constant coefficients Exact: $\mu(x) = 0.50000$; $\sigma(x) = 1.00000$					
Method 1	$0.49973 + 0.00027\,x$	1.00020	0.00008	0.70000	1.44870×10^{-14}
Method 2	0.50000	1.00020	0.00058	0.85000	1.35643×10^{-14}
ODE2	0.50000	1.00020	0.00058	0.85000	1.35643×10^{-14}
Quadratic coefficients Exact: $\mu(x) = 0.70000\,x + 0.30000\,x^2$; $\sigma(x) = 0.50000\,x + 0.10000\,x^2$					
Method 1	$0.62051\,x + 0.35905\,x^2$	$-0.01778 + 0.53598\,x + 0.08424\,x^2$	0.00087	0.50000	1.44796×10^{-7}
Method 2	$1.02213\,x$	$-0.01997 + 0.54009\,x + 0.08239\,x^2$	0.00087	0.40000	1.45359×10^{-7}
ODE2	$0.42731\,x + 0.45097\,x^2$	$-0.02019 + 0.54020\,x + 0.08250\,x^2$	0.00087	0.40000	1.56184×10^{-7}
Ornstein–Uhlenbeck Exact: $\mu(x) = -0.50000\,x$; $\sigma(x) = 0.30000$					
Method 1	$-0.50031 + 0.00031\,x$	0.29901	0.00087	0.40000	2.96264×10^{-16}
Method 2	-0.50000	0.29901	0.00087	0.40000	2.15839×10^{-16}
ODE2	$-0.50000\,x$	0.29901	0.00017	0.40000	2.15707×10^{-16}

Final MSE. Mean squared increment prediction error on the fitted data: $\frac{1}{N}\sum_{i=0}^{N-1}[\Delta X_i - \widehat{\mu}(X_i)\Delta t - \widehat{\sigma}(X_i)\Delta\widehat{W}_i]^2$, using each method's recovered noise and the final rounded coefficients. Values are the saved `increment_mse`, shown to six significant figures; these are not cross-validation errors.

Parameters. α is the ElasticNet regularization strength; ρ is its L1 mixing ratio. Each pair applies to the joint drift–diffusion fit and was selected by seven-fold cross-validation with the one-standard-error rule. All four experiments used 100,000 simulated samples.

Selection. Both polynomial libraries contain powers zero through five. The constant experiment uses the ρ grid: 0.70000, 0.85000, 0.95000, 1.00000.

Precision. Final recovered coefficients are rounded in code to five decimal places after model selection and refitting. Predictions, plots, and errors use these rounded models. Unrounded coefficients are saved for comparison. α and ρ are displayed to five decimal places; selection uses full precision. The exact functions define the simulated processes, and σ denotes diffusion amplitude, not variance.

Source. `outputs/<case>/ssisde_comparison.json` and `model_truth.json`, where `<case>` is `black_scholes`, `constant`, `quadratic`, or `ou`.